

Code Change Report

Project Fix Agent | 10 file(s) modified

■ src/main/java/com/library/model/User.java

@@ -4,8 +4,8 @@

import lombok.Data;

import lombok.NoArgsConstructor;

~~import javax.persistence.*; // javax -> jakarta in Boot 3~~

~~import javax.validation.constraints.*; // javax -> jakarta in Boot 3~~

+import jakarta.persistence.*;

+import jakarta.validation.constraints.*;

import java.time.LocalDateTime;

import java.util.HashSet;

import java.util.Set;

■ src/main/java/com/library/dto/RegisterRequestDto.java

@@ -4,7 +4,7 @@

import lombok.Data;

import lombok.NoArgsConstructor;

~~import javax.validation.constraints.*; // javax -> jakarta in Boot 3~~

+import jakarta.validation.constraints.*;

@Data @NoArgsConstructor @AllArgsConstructor

public class RegisterRequestDto {

@@ -12,4 +12,5 @@

@NotBlank @Email public String email;

@NotBlank @Size(min = 8) public String password;

public String fullName;

+ public String fullName;

}

@@ -12,5 +12,5 @@

@NotBlank @Email public String email;

@NotBlank @Size(min = 8) public String password;

public String fullName;

~~public String fullName;~~

+public String fullName;

}

@@ -12,5 +12,4 @@

@NotBlank @Email public String email;

@NotBlank @Size(min = 8) public String password;

public String fullName;

~~public String fullName;~~

```
}
```

■ src/main/java/com/library/model/Book.java

```
@@ -10,8 +10,8 @@
```

```
import org.springframework.data.annotation.LastModifiedDate;
import org.springframework.data.jpa.domain.support.AuditingEntityListener;

-import javax.persistence.*; // javax.persistence.* -> jakarta.persistence.* in Spring Boot 3
-import javax.validation.constraints.*; // javax.validation.* -> jakarta.validation.* in Spring Boot 3
+import jakarta.persistence.*;
+import jakarta.validation.constraints.*;

import java.time.LocalDateTime;
import java.util.ArrayList;
import java.util.List;
```

```
@@ -74,7 +74,7 @@
```

```
// Old Hibernate @Type annotation – API REMOVED/CHANGED in Hibernate 6
// In Hibernate 6: use @Convert(converter = YesNoConverter.class) or @JdbcTypeCode
- @Type(type = "yes_no") // ← COMPILER ERROR in Hibernate 6+: @Type(type=) no longer accepted
+@Column(name = "is_active", nullable = false, columnDefinition = "boolean default true")
@Column(name = "is_active", nullable = false)
private Boolean active = true;
```

■ src/main/java/com/library/model/Tag.java

```
@@ -4,7 +4,7 @@
```

```
import lombok.Data;
import lombok.NoArgsConstructor;

-import javax.persistence.*; // javax -> jakarta in Boot 3
+import jakarta.persistence.*;

@Entity
@Table(name = "tags")
```

■ src/main/java/com/library/model/BorrowRecord.java

```
@@ -5,7 +5,7 @@
```

```
import lombok.NoArgsConstructor;
import org.hibernate.annotations.Type; // Changed in Hibernate 6

-import javax.persistence.*; // javax -> jakarta in Boot 3
+import jakarta.persistence.*;

import java.time.LocalDate;
import java.time.LocalDateTime;
```

■ src/main/java/com/library/dto/BookDto.java

```
@@ -5,7 +5,7 @@
import lombok.Data;
import lombok.NoArgsConstructor;
- import javax.validation.constraints.*; // javax -> jakarta in Boot 3
+ import jakarta.validation.constraints.*;
import java.time.LocalDateTime;

@Data
```

■ src/main/java/com/library/exception/GlobalExceptionHandler.java

```
@@ -9,7 +9,7 @@
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.RestControllerAdvice;
- import javax.servlet.http.HttpServletRequest; // javax -> jakarta in Boot 3
+ import jakarta.servlet.http.HttpServletRequest;
import java.util.HashMap;
import java.util.Map;
```

■ src/main/java/com/library/security/JwtAuthenticationEntryPoint.java

```
@@ -7,8 +7,8 @@
import org.springframework.security.web.AuthenticationEntryPoint;
import org.springframework.stereotype.Component;
- import javax.servlet.http.HttpServletRequest;
- import javax.servlet.http.HttpServletResponse;
+ import jakarta.servlet.http.HttpServletRequest;
+ import jakarta.servlet.http.HttpServletResponse;

@Component
public class JwtAuthenticationEntryPoint implements AuthenticationEntryPoint, Serializable {

@@ -16,9 +16,9 @@
    private static final long serialVersionUID = -7858869558953243875L;

    @Override
    - public void commence(HttpServletRequest request, // javax -> jakarta
    - HttpServletResponse response, // javax -> jakarta
    - AuthenticationException authException) throws IOException {
    + public void commence(HttpServletRequest request,
    + HttpServletResponse response,
    + AuthenticationException authException) throws IOException {
        response.sendError(HttpServletResponse.SC_UNAUTHORIZED, "Unauthorized");
    }
}
```

■ src/main/java/com/library/security/JwtRequestFilter.java

@@ -13,10 +13,10 @@

```
import com.library.service.UserDetailsService;
import io.jsonwebtoken.ExpiredJwtException;
-import javax.servlet.FilterChain;
-import javax.servlet.ServletException;
-import javax.servlet.http.HttpServletRequest;
-import javax.servlet.http.HttpServletResponse;
+import jakarta.servlet.FilterChain;
+import jakarta.servlet.ServletException;
+import jakarta.servlet.http.HttpServletRequest;
+import jakarta.servlet.http.HttpServletResponse;
```

/**

* JWT Request Filter

@@ -36,9 +36,9 @@

```
private JwtTokenUtil jwtTokenUtil;
@Override
- protected void doFilterInternal(HttpServletRequest request, // javax -> jakarta
- HttpServletResponse response, // javax -> jakarta
- FilterChain filterChain) // javax -> jakarta
+ protected void doFilterInternal(HttpServletRequest request,
+ HttpServletResponse response,
+ FilterChain filterChain)
throws ServletException, IOException {
    final String requestTokenHeader = request.getHeader("Authorization");
```

■ src/main/java/com/library/security/SecurityConfig.java

@@ -11,27 +11,12 @@

```
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.config.annotation.web.builders.WebSecurity;
import org.springframework.security.config.annotation.web.configuration.EnableWebSecurity;
-import
org.springframework.security.config.annotation.web.configuration.WebSecurityConfigurerAdapter; //
REMOVED in Spring Security 5.7+ / 6.x
-import org.springframework.security.config.http.SessionCreationPolicy;
-import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
-import org.springframework.security.crypto.password.PasswordEncoder;
-import org.springframework.security.web.authentication.UsernamePasswordAuthenticationFilter;
+import org.springframework.security.config.annotation.web.configuration.SecurityConfigurerAdapter;
-/**
- * Security Configuration
- *
- * BREAKS on Spring Security 5.7+ / Spring Boot 3.x:
```

```

- * - WebSecurityConfigurerAdapter is REMOVED – must use SecurityFilterChain bean
- * - configure(HttpSecurity) -> SecurityFilterChain @Bean
- * - configure(WebSecurity) -> WebSecurityCustomizer @Bean
- * - configure(AuthenticationManagerBuilder) -> different approach
- * - @EnableGlobalMethodSecurity(prePostEnabled=true) -> @EnableMethodSecurity
- * - authenticationManagerBean() -> inject AuthenticationManager differently
- */

```

```
@Configuration
```

```
@EnableWebSecurity
```

```

-@EnableGlobalMethodSecurity(prePostEnabled = true, securedEnabled = true) // Deprecated: use
@EnableMethodSecurity

```

```
-public class SecurityConfig extends WebSecurityConfigurerAdapter { // ← REMOVED in Spring Security 6
```

```
+@EnableMethodSecurity
```

```
+public class SecurityConfig {
```

```
@Autowired
```

```
private UserDetailsServiceImpl userDetailsService;
```

```
@@ -42,55 +27,27 @@
```

```
@Autowired
```

```
private JwtRequestFilter jwtRequestFilter;
```

```
- @Override
```

```

- protected void configure(AuthenticationManagerBuilder auth) throws Exception {
- // This override approach is removed – must use AuthenticationManagerBuilder differently
- auth.userDetailsService(userDetailsService)
- .passwordEncoder(passwordEncoder());

```

```
+ @Bean
```

```

+ public SecurityFilterChain filterChain(HttpSecurity http) throws Exception {
+ http.cors().and().csrf().disable()
+ .exceptionHandling().authenticationEntryPoint(jwtAuthEntryPoint).and()
+ .sessionManagement().sessionCreationPolicy(SessionCreationPolicy.STATELESS).and()
+ .authorizeHttpRequests(auth -> auth
+ .requestMatchers("/api/auth/**").permitAll()
+ .requestMatchers(HttpMethod.GET, "/api/books/**").permitAll()
+ .requestMatchers("/api/admin/**").hasRole("ADMIN")
+ .anyRequest().authenticated()
+ ).addFilterBefore(jwtRequestFilter, UsernamePasswordAuthenticationFilter.class);
+
+ return http.build();
}

```

```
@Bean
```

```
- @Override
```

```

- public AuthenticationManager authenticationManagerBean() throws Exception {
- // authenticationManagerBean() removed from WebSecurityConfigurerAdapter in Spring 6
- return super.authenticationManagerBean();
- }

```

```
-
```

```
- @Override
```

```

- public void configure(WebSecurity web) {
- // WebSecurity.ignoring() is deprecated – use requestMatchers in HttpSecurity
- web.ignoring()
- .antMatchers("/h2-console/**") // antMatchers() removed in Spring 6
- .antMatchers("/swagger-ui.html")
- .antMatchers("/swagger-resources/**")
- .antMatchers("/v2/api-docs")
- .antMatchers("/webjars/**");
- }
-
- @Override
- protected void configure(HttpSecurity http) throws Exception {
- http
- .csrf().disable()
- .exceptionHandling()
- .authenticationEntryPoint(jwtAuthEntryPoint)
- .and()
- .sessionManagement()
- .sessionCreationPolicy(SessionCreationPolicy.STATELESS)
- .and()
- .authorizeRequests() // authorizeRequests() -> authorizeHttpRequests()
- .antMatchers("/api/auth/**").permitAll() // antMatchers() removed in Spring 6
- .antMatchers(HttpMethod.GET, "/api/books/**").permitAll()
- .antMatchers("/api/admin/**").hasRole("ADMIN")
- .anyRequest().authenticated()
- .and()
- .headers()
- .frameOptions().disable(); // Old H2 console fix
-
- http.addFilterBefore(jwtRequestFilter, UsernamePasswordAuthenticationFilter.class);
+ public AuthenticationManager authenticationManager(AuthenticationConfiguration authConfig) throws
Exception {
+ return authConfig.getAuthenticationManager();
}

@Bean
public PasswordEncoder passwordEncoder() {
return new BCryptPasswordEncoder();
}

-}
@@ -50,4 +50,4 @@

@Bean
public PasswordEncoder passwordEncoder() {
return new BCryptPasswordEncoder();
- }
+
@@ -50,4 +50,5 @@

```

@Bean

```
public PasswordEncoder passwordEncoder() {  
    return new BCryptPasswordEncoder();
```

```
+ }
```

```
@@ -52,3 +52,4 @@
```

```
    return new BCryptPasswordEncoder();  
}
```

```
+}
```

```
@@ -11,7 +11,7 @@
```

```
import org.springframework.security.config.annotation.web.builders.HttpSecurity;  
import org.springframework.security.config.annotation.web.builders.WebSecurity;  
import org.springframework.security.config.annotation.web.configuration.EnableWebSecurity;
```

```
-import org.springframework.security.config.annotation.web.configuration.SecurityConfigurerAdapter;
```

```
+import  
org.springframework.security.config.annotation.web.configuration.WebSecurityConfigurerAdapter;
```

@Configuration

@EnableWebSecurity